

```
<!DOCTYPE html>

<html>

<head>

  <title>JavaScript Fundamentals</title>

</head>

<body>


<h2>JavaScript Fundamentals Demo</h2>


<button onclick="handleClick()">Click Me</button>


<script>


  // INTRODUCTION: JS in HTML (Internal JS used here)
  console.log("JavaScript is running inside HTML");


  // VARIABLES
  var num1 = 10;
  var num2 = 5;


  // OPERATIONS
  var sum = num1 + num2;
  var diff = num1 - num2;
```

```
console.log("Sum:", sum);  
console.log("Difference:", diff);
```

```
// CONDITIONAL STATEMENTS
```

```
if (sum > 10) {  
    console.log("Sum is greater than 10");  
} else {  
    console.log("Sum is 10 or less");  
}
```

```
// FUNCTION DECLARATION
```

```
function greet(name) {  
    return "Hello " + name;  
}
```

```
console.log(greet("Student"));
```

```
// FUNCTION EXPRESSION
```

```
var multiply = function(a, b) {  
    return a * b;  
};
```

```
console.log("Multiply:", multiply(3, 4));
```

```
// HOISTING
```

```
console.log(x); // undefined due to hoisting
```

```
var x = 20;
```

```
sayHi(); // works before declaration
```

```
function sayHi() {
```

```
    console.log("This is hoisting example");
```

```
}
```

```
// INPUT & OUTPUT
```

```
var userName = prompt("Enter your name:");
```

```
alert("Welcome " + userName);
```

```
console.log("User entered:", userName);
```

```
// EVENT HANDLING
```

```
function handleClick() {
```

```
    alert("Button Clicked!");
```

```
var result = num1 + num2;
```

```
if (result > 10) {  
    alert("Result is greater than 10");  
} else {  
    alert("Result is small");  
}
```

```
    alert("Multiplication: " + multiply(num1, num2));  
}
```

```
</script>
```

```
</body>
```

```
</html>
```